

Skills Summary

Complex Solution Sets

Use this reference while completing your worksheet or when studying.

Skill 1 — Square Roots of Negative Numbers

Rule: $\sqrt{-k} = i\sqrt{k}$ for $k > 0$, and a square has *two* roots: $x^2 = -k$ gives $x = \pm i\sqrt{k}$.

Isolate the square first; divide out any leading coefficient before taking roots.

Example: Solve $x^2 + 36 = 0$

Step 1. The square is already alone on one side after one move, so there is no need for the quadratic formula — isolate and take both roots.

Step 2. $x^2 = -36$, and the two square roots of -36 are $6i$ and its negative.

$$\Rightarrow x = 6i \text{ or } x = -6i$$

Try it: Solve $2x^2 + 98 = 0$

check: $-x \text{ or } x = x$

Skill 2 — When the Discriminant Is Negative

Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When $b^2 - 4ac < 0$ the two solutions are non-real conjugates. Divide *both* parts of the numerator by $2a$.

Example: Solve $x^2 - 6x + 25 = 0$

Step 1. There is no factor pair for 25 that sums to -6 , so go to the quadratic formula and read the discriminant first.

Step 2. The discriminant is $36 - 100 = -64$, so the numerator is $6 \pm 8i$; dividing both parts by 2 gives real part 3 and imaginary part 4.

$$\Rightarrow x = 3 \pm 4i$$

Try it: Solve $x^2 + 2x + 26 = 0$

check: $1 - x = x$

Skill 3 — All Solutions of a Higher Degree

Rule: an equation of degree n has exactly n solutions, counted with multiplicity. Count what you have before you stop.

Factor into quadratics, then solve each one — a factor like $x^2 + 4$ contributes a non-real pair, and it is the factor most often abandoned.

Example: Solve $x^4 - 16 = 0$

Step 1. Degree four means four solutions, so factoring as far as possible is the way to find all of them.

Step 2. $(x^2 - 4)(x^2 + 4) = 0$: the first factor gives the real pair, and the second gives $x^2 = -4$, whose roots are imaginary.

$$\Rightarrow x = 2, -2, 2i, -2i$$

Try it: Solve $x^4 - 5x^2 - 36 = 0$

check: $x = 2, -2, 3i, -3i$

Skill 4 — Checking and Reporting the Full Set

To check a solution: substitute it in and simplify. If the expression comes out 0, the value is a solution.

Conjugate pairs: when every coefficient is real, non-real solutions come in pairs $a + bi$ and $a - bi$ — so finding one hands you the other.

Example: Is $x = 2 - i$ a solution of $x^2 - 4x + 5 = 0$?

Step 1. Checking a proposed solution means substituting it and simplifying, not re-solving the equation.

Step 2. $(2 - i)^2 = 3 - 4i$ and $-4(2 - i) = -8 + 4i$; adding those with 5, the imaginary parts cancel and the real parts give $3 - 8 + 5$.

$$\Rightarrow 0, \text{ so yes — and } 2 + i \text{ is the other solution.}$$

Try it: Substitute $x = 3 + i$ into $x^2 - 6x + 10$.

check: 0